

DHARSAN KESAVAN

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EDUCATION

Arizona State University, Tempe, AZ

Aug 2026 – Expected May 2030

Dual B.S., Biological Sciences (Biomedical Sciences) and AI in Business. Barrett, The Honors College.

- McKenna Life Sciences, Business and Entrepreneurship (MLSBE), **inaugural cohort**. A 30-student program run jointly by W. P. Carey and The College: two bachelor's degrees plus an Entrepreneurship and Innovation certificate in four years.
- 2026 AI Scholar cohort. Member, Sun Devil Satellite Laboratory. Entered with 51 credit hours from AP exams.

BASIS Chandler, Chandler, AZ

Aug 2022 – May 2026

- AP Scholar with Distinction. Scores of 5 in AP Biology, AP Chemistry, and AP Physics 1. SAT 1530.

ENGINEERING AND VENTURES

VivaMed BioPharma, Paid Technical Intern

Aug 2026 – Present

Computational drug-candidate screening. Python, Pydantic, Polars, RDKit, Postgres.

- Primary engineer on a computational screening platform for drug candidates: wrote most of its screening engines, the scoring and red-team review stages, and the orchestration that runs them end to end, so each run yields a reproducible, provenance-tracked evidence dossier. More than 300 test files gate every change.
- Built the data layer's safeguards: loaders and reconciliation over a 500 GB Postgres research warehouse (DepMap, ClinVar, FAERS), and a licence gate that stops any engine from reading a source whose terms bar that use. It is on by default, and every bypass needs a recorded reason.
- Merged the platform with the CTO's discovery-engine codebase into one pipeline of 19 cost-ordered gates, so cheap checks eliminate a candidate before expensive ones run, and pulled the scoring logic both had duplicated into one shared package, resolving a constant the two had silently disagreed on.
- Hardened the CTO-led engine suite with verification gates, a default-deny network egress allowlist, database writes that fail loudly instead of silently, and an auto-generated project status site.

Cognifer Labs, Founder

Mar 2026 – Present

Avorio, a native spaced-repetition learning app for macOS and iOS. avorio.ai

- Shipped Avorio to the App Store — version 1.0 approved July 2026, and 1.1 approved and released the same day it was submitted — with the Mac build available as a signed, notarized direct download that updates itself.
- Designed one core so every platform behaves identically: a single engine handles scheduling, storage, Anki import and export, gamification, and AI routing, and both native UIs call straight into it.
- Made switching from Anki lossless: a deck keeps its full history, media, cloze, and image occlusion behind a review step, then re-fits the learner's own forgetting-curve parameters from that history.
- Shipped an AI layer with free on-device paths, so cost is never the barrier: document-to-cards with diagrams, per-card explanations, and a tutor aimed at whatever the learner keeps missing.
- Ran the backend and business model: free flashcard core, paid tiers for cloud AI and sync, checkout live on web and both app stores, eight languages, a test suite gating every change, and Android in progress on the same core.

BioPath, Founder and Sole Engineer

2025 – Present

Drug interaction checker and compound body-impact analysis. biopath.space

- Built a computational pharmacology tool that answers what a compound will do inside a body and whether it is safe with what someone already takes, starting from a drug name, plant photo, plant name, or pill imprint.
- Connected ten scientific databases (PubChem, ChEMBL, Reactome, Open Targets, DGIdb, OpenFDA, RxNorm, PlantNet, Dr. Duke's, Drugs.com) into one pipeline: identity, binding targets, 200+ pathways, organ impact, then interaction, dosage, and pregnancy safety.
- Trained a Morgan-fingerprint CNN and MPNN on BindingDB to predict binding affinity across roughly 700 human proteins for compounds with no measured data, cross-checked against measured ChEMBL targets with disagreement flagged rather than hidden. Tanimoto similarity at 0.70 carries side effects and interaction risk over from analogous known drugs.

AlóJefe, Full-Stack Developer (client)

2025 – 2026

- Built the site and backend for a bilingual AI phone assistant for solo contractors: it answers the calls an owner cannot take, qualifies the job, books an estimate with drive-time routing, and texts back a summary in English or Spanish. Next.js on Vercel.

CLINICAL EXPERIENCE

Banner Health, Patient-Facing Volunteer

2024 – Present · 175 hours

- Assisted patients and clinical staff across inpatient and outpatient departments at one of the largest nonprofit health systems in the country. Built a front-desk data system that cut the time it took to log and retrieve patient information.

Desert Kidney Associates, Physician Shadowing, Nephrology	2024 – 2026 · 120 hours
Shadowed Dr. Prashant Kolar through inpatient rounds, outpatient consultations, and dialysis management: CKD staging, fluid and electrolyte management, and how diabetes and hypertension become renal outcomes, including patient education on lifestyle change.	
ROC Physical Therapy, Clinical Shadowing	2023 · 80 hours
Observed outpatient rehabilitation with physical therapist Salman Ashraf: gait and movement assessment, therapeutic exercise progression, manual therapy, and the arc from injury back to function.	
Emergency Medical Technician, training in progress	2026

RESEARCH AND SCHOLARLY WRITING

"Non-Pharmacological Therapies for Alzheimer's Disease: MST and CST," <i>literature review</i>	Sept 2025
Compared Mnemonic Strategy Training and Cognitive Stimulation Therapy for <i>Mindful Mental Health</i> on Medium across the neuroimaging evidence and published effect sizes; showed why the two cannot be compared head-to-head (different outcome measures and patient populations) and argued they are complementary.	

LEADERSHIP AND TEACHING

Flare Review, Director of Operations	Oct 2025 – Aug 2026
Ran operations for a student-led video editing nonprofit: scheduling, project intake, and hand-off between editors.	
Schoolhouse.world, Senior Tutor	Nov 2023 – Jun 2026
Tutored one on one against what each student was actually getting wrong; recruited and onboarded new tutors.	
Volunteer Teacher, student meal program	2023 – 2025
Raised \$7,000 for student meal programs and taught in the classrooms that money paid for.	
Arizona Tamil School, Teaching Assistant	10+ years
Taught language and culture classes for a decade and coordinated the annual performances.	

OTHER SOFTWARE

Agent Bridge. A Tauri desktop app in Rust and React that drives Claude Code, Codex, and Cursor from one shell over the Agent Client Protocol. One canonical config projects into each tool's native format and round-trips back unchanged, verified by identity tests. 10,400 lines, 117 tests. Backglass. An open-source (MIT), local-first commitment ledger in Python (184 modules, 18 source connectors, 3,200+ tests). It reads mail, messages, calendars, Canvas, and notes on a schedule and extracts typed records, each linked to its source: what you owe, what is owed to you, deadlines, and goal checkpoints. No chat interface and no vector database, on purpose: the questions are known in advance, so the work belongs at ingest in plain SQL. It serves a versioned personal fact base to other AI agents over MCP, and a zero-cost pre-filter removed 10.9% of 3,687 real messages with zero disagreement against the model verdicts it replaced. Tradgent. A personal investing companion in Python (66 modules, 2,300+ tests) that tracks contributions, allocation drift, tax lots, and wash sales, with no broker write path in its core, enforced by a test. A separately fenced signals lane scores news, SEC filings, and Federal Reserve and Treasury feeds into a rating per instrument and trades on paper only, dry by default. Misinformation contagion simulators. Two agent-based models of a claim spreading through a synthetic Phoenix. The C++ and SFML build runs 10,000 agents over 690 steps on an SEDPNR state model, contact weighted by shared school, workplace, age, and ethnicity, with false claims carrying the six-fold transmission advantage Vosoughi and colleagues measured in <i>Science</i> (2018). The Python build adds System 1 and System 2 reasoning and biased media diets.	
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HONORS AND CERTIFICATIONS

Honors	Best Delegate, Model United Nations. One gold and three silver medals, Academic Decathlon state competition.
Certifications	Introductory Human Physiology, Duke University (Aug 2025, credential 3SYLO9PVP12V). Machine Learning in Healthcare, Northeastern University. OSHA Bloodborne Pathogens and HIPAA for healthcare workers.

SKILLS

Biomedical	Pharmacology: drug mechanisms, receptor binding, ADME, drug-drug interactions. Biomedical data: PubChem, ChEMBL, BindingDB, Reactome, Open Targets, DGIdb, OpenFDA, RxNorm, DrugBank, DepMap, ClinVar, FAERS. Renal physiology. Literature review and effect-size interpretation.
Programming	Rust, Swift/SwiftUI, Kotlin/Jetpack Compose, Python (Pydantic, Polars), TypeScript, C++. UniFFI, rusqlite, React, Next.js, Tauri.
Machine learning	Morgan-fingerprint CNN and MPNN training on BindingDB, cross-validation against measured data, Tanimoto similarity, RDKit, LLM integration through Anthropic, OpenAI, and Ollama, MCP servers.
Infrastructure	Supabase and Postgres, SQLite in WAL mode, Deno edge functions, Vercel, Stripe, RevenueCat and StoreKit, GitHub Actions CI.
Business	Pricing and tiering, App Store review, notarization, localization, freelance delivery, fundraising.